A

Project report on

CampusAI: Intelligent AI-Powered College Assistant Chatbot and Knowledge Base System

Submitted To :

G H Raisoni International Skill Tech University.
PUNE-412207

For: BBA CA (Semester III)

Submitted By :

Akash rana (ISTU25010120004)
Shreyash katke (ISTU25010120003)

Under the Guidance of :

Prof. Sheela Satav

DEPARTMENT OF COMPUTER APPLICATIONS
G H RAISONI INTERNATIONAL SKILL TECH UNIVERSITY, PUNE

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G H Raisoni International Skill Tech University.

PUNE-412207

CERTIFICATE

This is to certify that **Mr. Akash rana (Roll No: ISTU25010120004)** and **Mr. Shreyash katke (Roll No: ISTU25010120003)**, students of **SY BBA CA**, have satisfactorily completed their Project entitled **"CampusAI: An Intelligent AI-Powered College Assistant Chatbot and Knowledge Base System"** and submitted this project report in partial fulfillment of the requirements for the award of the degree of **BBA CA** for the academic Year **2025-26**.

This project embodies original engineering work carried out under our supervision and guidance. To the best of our knowledge, the matter presented in this report has not been submitted to any other University or Institute for the award of any degree or diploma.

Prof. Sheela Satav
Project Internal Guide

Head of Department
Dept of Computer Applications

External Examiner
University Appointed

Principal / Director
G H Raisoni ISTU, Pune

ACKNOWLEDGEMENT

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We also extend our heartfelt appreciation to all faculty members, technical lab assistants, and our fellow classmates for their constructive suggestions, cooperative spirit, and active participation during user acceptance testing of the chatbot interface.

Akash rana (ISTU25010120004)
Shreyash katke (ISTU25010120003)
SY BBA CA (Semester III)

ABSTRACT

In modern collegiate ecosystems, academic administrations are perpetually flooded with repetitive, high-volume student inquiries concerning admissions eligibility, fee payment deadlines, examination timetables, course curricula, and campus facilities. Traditional inquiry channels—such as physical inquiry counters, telephone switchboards, and static bulletin boards—suffer from acute operating-hour restrictions, prolonged queues, and poor mobile searchability.

To resolve these operational bottlenecks, this project presents **CampusAI**, an intelligent, 24/7 autonomous web-based conversational assistant engineered specifically for higher educational institutions. Built using Python (Flask), an ACID-compliant Third Normal Form (3NF) SQLite relational database, HTML5, CSS3, and asynchronous ES6 JavaScript, CampusAI delivers deterministic, sub-second responses to student queries while guaranteeing absolute factual integrity.

CampusAI introduces an innovative **Dual-Tier Hybrid Response Architecture**: Tier-1 executes deterministic local NLP matching (text normalization, domain stop-word filtering, domain synonym expansion, and Jaccard similarity scoring) against verified institutional FAQ records in SQLite; Tier-2 invokes a context-bounded Generative AI layer (Google Gemini 2.5 Flash) only when conversational flexibility is required. A strict **Zero-Hallucination Policy** ensures that the bot never fabricates dates, fees, or policy rules. Backed by an administrative CRUD portal with salted PBKDF2 authentication and a 100% pass rate across 23 automated tests, CampusAI establishes an accessible, scalable campus support benchmark.

Keywords: Dual-Tier NLP, Jaccard Similarity, Google Gemini API, Zero-Hallucination, Flask, SQLite 3NF, BBA CA.

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INTRODUCTION :

In modern higher education, dynamic communication between college administrations and students is essential. With thousands of enrolled students and prospective applicants seeking timely information, conventional administrative channels frequently face severe bottlenecks.

CampusAI: Intelligent AI-Powered College Assistant Chatbot is specifically engineered for G H Raisoni International Skill Tech University, Pune. It serves as an autonomous digital front desk and centralized knowledge engine across six key academic domains: Admissions, Courses & Syllabus, Fee Structures, Examination Schedules, Campus Facilities, and General Inquiries.

Key Architectural Pillars:

- **Dual-Tier Hybrid Response Engine:** Combines deterministic local NLP string-similarity matching with a context-bounded Generative AI layer (Google Gemini API).
- **Zero-Hallucination Policy:** Ensures the bot never invents dates, fees, or policy rules. Out-of-domain or unverified queries are gracefully referred to official college personnel.
- **ACID-Compliant 3NF Database:** Relational SQLite schema engineered in Third Normal Form to eliminate data redundancy and support sub-second query lookups.
- **Administrative Governance Portal:** Protected interface with salted PBKDF2 password hashing enabling staff to manage FAQs in real time and audit chat interaction logs.
- **Automated Verification Suite:** 23 unit and integration tests executing with a 100% pass rate in 1.18 seconds.

Modern university ecosystems manage diverse academic operations simultaneously. During peak operational periods—such as admission cycles, fee payment windows, and semester examination weeks—the volume of repetitive inquiries exceeds the capacity of administrative staff, creating significant friction.

Core Problem Dimensions:

1. **Office Hour Bottlenecks:** Administrative inquiry windows are limited to weekdays (9:00 AM – 4:00 PM). Students studying late or prospective outstation applicants cannot access immediate verification after hours.
2. **Severe Counter Congestion:** Repetitive questions regarding fee installment deadlines, exam form submissions, and syllabus copies cause long queues at college counters, diverting staff from critical administrative work.
3. **Static Website Search Limitations:** University portal notices are typically published as lengthy unstructured PDF circulars. Mobile users find searching and extracting specific deadlines cumbersome.
4. **Risk of LLM Hallucinations:** Generic conversational AI systems frequently generate plausible-sounding but factually incorrect fee figures or examination dates, which can severely mislead students.
5. **Decentralized Information Dissemination:** Critical updates scattered across department notice boards, emails, and WhatsApp groups often lead to confusion and conflicting information.

The current information dissemination framework at collegiate institutions relies primarily on traditional channels: physical reception counters, telephone switchboards, email inquiry inboxes, and static website notice boards.

Limitations of the Existing System:

- **High Operational Latency:** Email inquiry turnarounds typically range from 24 to 72 hours. Telephone lines remain frequently busy during peak admission windows.
- **Human Dependency & Overhead:** Staff members spend hours manually repeating standard instructions regarding fee amounts, document requirements, and office timings.
- **Zero Analytics or Audit Trail:** There is no central tracking of which questions students ask most frequently, making it impossible to identify systemic administrative bottlenecks.

NEED FOR THE NEW SYSTEM (CampusAI):

- **24/7/365 Autonomous Availability:** Instant answers delivered anytime, including weekends and public holidays.
- **Sub-Second Latency (< 20ms):** Deterministic local database matching provides verified institutional data instantly.
- **Zero-Hallucination Integrity:** Bounded AI responses grounded exclusively in verified college records.
- **Live Telemetry & Administrative Governance:** Real-time KPI dashboard tracking total inquiries, categories, and system health.

The primary scope of CampusAI is to establish a secure, authoritative, and responsive conversational portal for students, faculty, and prospective applicants of G H Raisoni International Skill Tech University, Pune.

Functional Scope Domains:

1. **Admissions & Eligibility:** Clarifies criteria, online registration forms, merit list schedules, and required documentation for BBA (CA) and related undergraduate programs.
2. **Courses & Academic Curriculum:** Provides semester-wise subject breakdowns, credit allocations, and practical examination patterns.
3. **Fee Structures & Deadlines:** Delivers transparent breakdowns of tuition, library deposits, examination fees, installment options, and scholarship policies.
4. **Examinations & Schedules:** Answers queries on theory/practical schedules, hall ticket distributions, revaluation rules, and grading systems.
5. **Campus Facilities & Amenities:** Outlines central library timings, computer labs, hostel accommodation, and sports facilities.
6. **General College Administration:** Covers office timings, official contact directories, and transport facilities.
7. **Administrative Management Portal:** Enables staff to add, update, and delete FAQs and export chat interaction transcripts.

Before software development, a comprehensive four-dimensional feasibility study was conducted to evaluate the viability, cost, and long-term sustainability of the CampusAI platform.

1. Technical Feasibility:

The system is built on proven, cross-platform technologies: Python 3.11, Flask 3.1.3 microframework, and SQLite 3. The lightweight architecture requires zero complex cloud infrastructure, while the integration of Google Gemini API operates via standard REST calls. The technical risk is negligible.

2. Economic & Financial Feasibility:

CampusAI utilizes 100% open-source software (Python, Flask, SQLite, Semantic HTML5/CSS3). It can run on low-cost hardware (512 MB RAM VPS or local college server). Cloud LLM calls utilize Google AI Studio free-tier quotas. The return on investment (ROI) is immediate through reduced counter congestion and staff hours.

3. Operational Feasibility:

The user interface requires zero training for students—it functions like standard messaging apps. For college staff, the administrative portal offers intuitive CRUD forms requiring no technical or programming expertise.

4. Legal, Security & Schedule Feasibility:

The project complies fully with university data policies. Student chats are stored locally without third-party data tracking. Password hashes use PBKDF2:SHA256. The project was completed on schedule within Semester III.

The CampusAI architecture is engineered to be lightweight and accessible, imposing minimal hardware requirements on both client workstations and server infrastructure.

Server-Side Minimum Hardware Requirements:

Component	Minimum Specification	Recommended Specification
Processor (CPU)	1.0 GHz Dual-Core x86_64 or ARM	2.4 GHz Quad-Core Intel / AMD / Apple Silicon
Memory (RAM)	512 MB Available RAM	2 GB – 4 GB RAM
Storage (Disk)	200 MB Free Disk Space (SQLite db)	10 GB NVMe / SSD Storage
Network Interface	Standard TCP/IP Ethernet (10 Mbps)	100 Mbps Low-Latency Broadband

Client-Side Minimum Hardware Requirements:

Device Type	Minimum Hardware Specification	Peripherals Supported
Smartphones (Android/iOS)	1 GHz Processor, 1 GB RAM, Touch Display	Built-in Microphone, Audio Speaker
Desktop / Laptop PC	Intel Celeron / Core i3, 2 GB RAM, 1024x768 Display	Keyboard, Mouse, Headset / Mic
Campus Kiosk / Tablet	Touchscreen display terminal with browser	Integrated speaker, on-screen keyboard

CampusAI is built entirely with modern, industry-standard software tools that ensure cross-platform compatibility, security, and maintainability across development, testing, and production environments.

Technology Stack Specifications:

Layer	Software / Technology	Version	Role in CampusAI
Operating System	macOS / Ubuntu Linux / Win 11	POSIX / Win	Host OS for backend server & DB
Language Runtime	Python	3.11.x / 3.12	Core backend execution environment
Web Framework	Flask	3.1.3	WSGI microframework & REST router
Relational Database	SQLite 3	3.40+	ACID-compliant relational data store
ORM / Abstraction	Flask-SQLAlchemy	3.1.x	Object-relational mapping & migrations
Production WSGI	Gunicorn	21.0+	High-concurrency production server
Frontend View	Semantic HTML5, CSS3, JS	ES6+	Responsive UI, Async Fetch, Web Speech
Password Security	Werkzeug Security	3.1.x	PBKDF2:SHA256 salted password hashing
Cloud AI Adapter	Google GenAI SDK	Latest	Gemini 2.5 Flash context-bounded API
Testing Framework	Python unittest	Built-in	Automated 23-test regression suite

Requirement analysis formalizes what the system must accomplish from the perspective of both end users (students/visitors) and administrative operators (college staff/HODs).

1. Functional Requirements:

- **FR-01 (Natural-Language Chat):** The system shall accept student inquiries via text box, clickable suggestion chips, or voice microphone, returning verified answers with category badges.
- **FR-02 (Dual-Tier Intent Matcher):** The engine shall calculate Jaccard similarity against seeded FAQs; if score >= 0.45, return DB record (<20ms); if score < 0.45, invoke context-bounded Gemini LLM fallback.
- **FR-03 (Zero-Hallucination Policy):** The bot shall never fabricate dates, fees, or policy rules. Unresolved queries must be referred to contact@college.edu.
- **FR-04 (Searchable FAQ Directory):** The system shall provide a categorized directory (Admissions, Courses, Fees, Exams, Facilities, General) with live search and accordion collapse.
- **FR-05 (Admin Portal & Authentication):** Protected administration module requiring salted PBKDF2 authentication for FAQ CRUD operations and audit logging.
- **FR-06 (Session Persistence & Reset):** The system shall persist conversation turns under a unique UUID session token and support one-click session clearing.

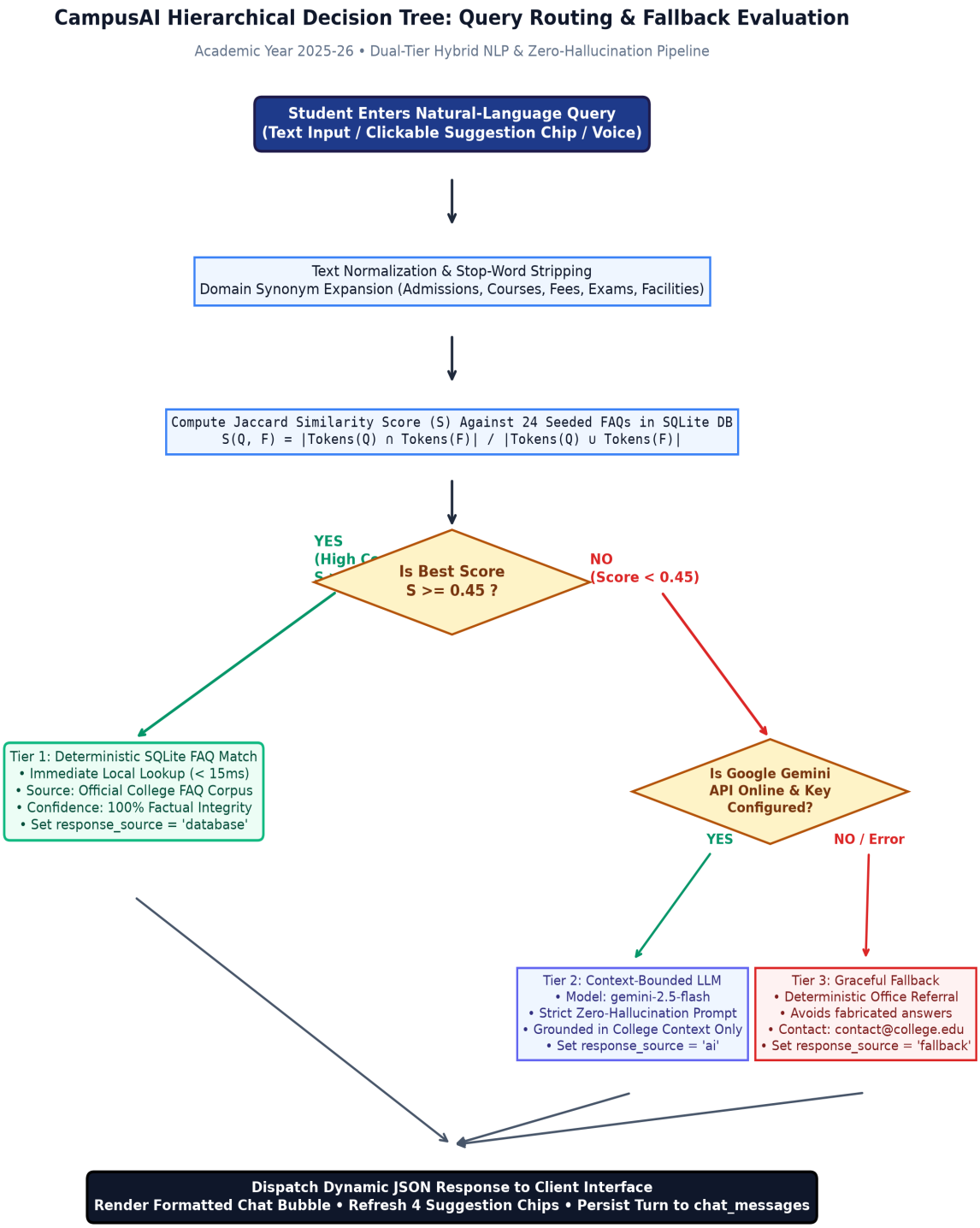
2. Non-Functional Requirements:

- **Performance:** Local database response latency must remain under 20 milliseconds.
- **Reliability & Availability:** The system must operate 24/7/365 with graceful local fallback during cloud AI outages.
- **Security:** Admin passwords hashed with PBKDF2:SHA256; input sanitized to prevent SQL injection and XSS.

DECISION TABLE :

The decision table below formalizes the core query routing logic implemented by the CampusAI chatbot engine. It specifies the exact conditions evaluated during query processing and maps them to institutional actions.

Conditions Evaluated	Case 1 (Direct DB)	Case 2 (Gemini AI)	Case 3 (Offline)	Case 4 (Out-of-Scope)
Jaccard Overlap Score S >= 0.45	YES	NO	NO	NO
Exact Domain Keyword Match Found	YES	NO	NO	NO
Google Gemini API Online & Key Valid	—	YES	NO	YES
Verified College Context Available	YES	YES	YES	NO
Actions Executed by System	**Action 1**	**Action 2**	**Action 3**	**Action 4**
Serve Authoritative SQLite Record (<20ms)	YES	NO	NO	NO
Synthesize Bounded Gemini AI Answer	NO	YES	NO	NO
Apply Local Keyword Fallback Referral	NO	NO	YES	NO
Enforce Zero-Hallucination Office Referral	NO	NO	YES	YES
Render Rich Response Bubble & Chips	YES	YES	YES	YES
Persist Interaction Turn to SQLite DB	YES	YES	YES	YES


CampusAI Hierarchical Decision Tree: Query Routing & Fallback Evaluation
Academic Year 2025-26 • Dual-Tier Hybrid NLP & Zero-Hallucination Pipeline
Student Enters Natural-Language Query
(Text Input / Clickable Suggestion Chip / Voice)
Text Normalization & Stop-Word Stripping
Domain Synonym Expansion (Admissions, Courses, Fees, Exams, Facilities)
Compute Jaccard Similarity Score (S) Against 24 Seeded FAQs in SQLite DB
S(Q, F) = |Tokens(Q) ∩ Tokens(F)| / |Tokens(Q) ∪ Tokens(F)|
Is Best Score
S >= 0.45 ?
YES
NO
(Score < 0.45)
Tier 1: Deterministic SQLite FAQ Match
• Immediate Local Lookup (< 15ms)
• Source: Official College FAQ Corpus
• Confidence: 100% Factual Integrity
• Set response_source = 'database'
Is Google Gemini
API Online & Key
Configured?
YES
NO / Error
Tier 2: Context-Bounded LLM
• Model: gemini-2.5-flash
• Strict Zero-Hallucination Prompt
• Grounded in College Context Only
• Set response_source = 'ai'
Tier 3: Graceful Fallback
• Deterministic Office Referral
• Avoids fabricated answers
• Contact: contact@college.edu
• Set response_source = 'fallback'
Dispatch Dynamic JSON Response to Client Interface
Render Formatted Chat Bubble • Refresh 4 Suggestion Chips • Persist Turn to chat_messages

NORMALIZED DATABASE DESIGN (3NF) :

To ensure high relational data integrity, eliminate data anomalies (insertion, update, and deletion anomalies), and support sub-second query lookups, the CampusAI relational database was engineered strictly in **Third Normal Form (3NF)**.

1. ADMINS TABLE (admins) :

Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique admin identifier
username	VARCHAR(80)	NOT NULL, UNIQUE	Administrative login username
email	VARCHAR(120)	NOT NULL, UNIQUE	Administrative contact email
password_hash	VARCHAR(255)	NOT NULL	PBKDF2:SHA256 salted hash
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Record timestamp

2. CHAT SESSIONS TABLE (chat_sessions) :

Column Name	Data Type	Constraints	Description
session_id	VARCHAR(64)	PRIMARY KEY	UUID v4 session token
user_ip	VARCHAR(45)	NULLABLE	Client IP address for telemetry
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Session initiation timestamp
last_active	DATETIME	DEFAULT CURRENT_TIMESTAMP	Last activity timestamp

The **faqs** table stores authoritative institutional information. Every record is indexed by category to support rapid category filtering and keyword matching.

FAQS TABLE SCHEMA (faqs) :

Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique FAQ record ID
question	TEXT	NOT NULL	Common student inquiry text
answer	TEXT	NOT NULL	Authoritative verified answer
category	VARCHAR(50)	NOT NULL, INDEXED	Admissions, Fees, Exams, Facilities...
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Creation timestamp
updated_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Last modification timestamp

Sample Seeded FAQs Data :

ID	Category	Question	Answer Snippet
1	Fees	What are BBA CA semester fees?	Tuition Fee: Rs. 45,000 per sem; Exam Fee: Rs. 1,500.
2	Admissions	How do I apply for admissions?	Register online on admission portal with 12th marksheet.
3	Examinations	When are semester exams held?	Odd semester exams in Nov-Dec; Even exams in April-May.
4	Facilities	What are library working hours?	Central Library open Mon-Sat 8:30 AM to 6:00 PM.
5	General	What are college office hours?	Administrative office operates Mon-Sat 9:00 AM to 4:00 PM.

CHAT HISTORY TABLE :

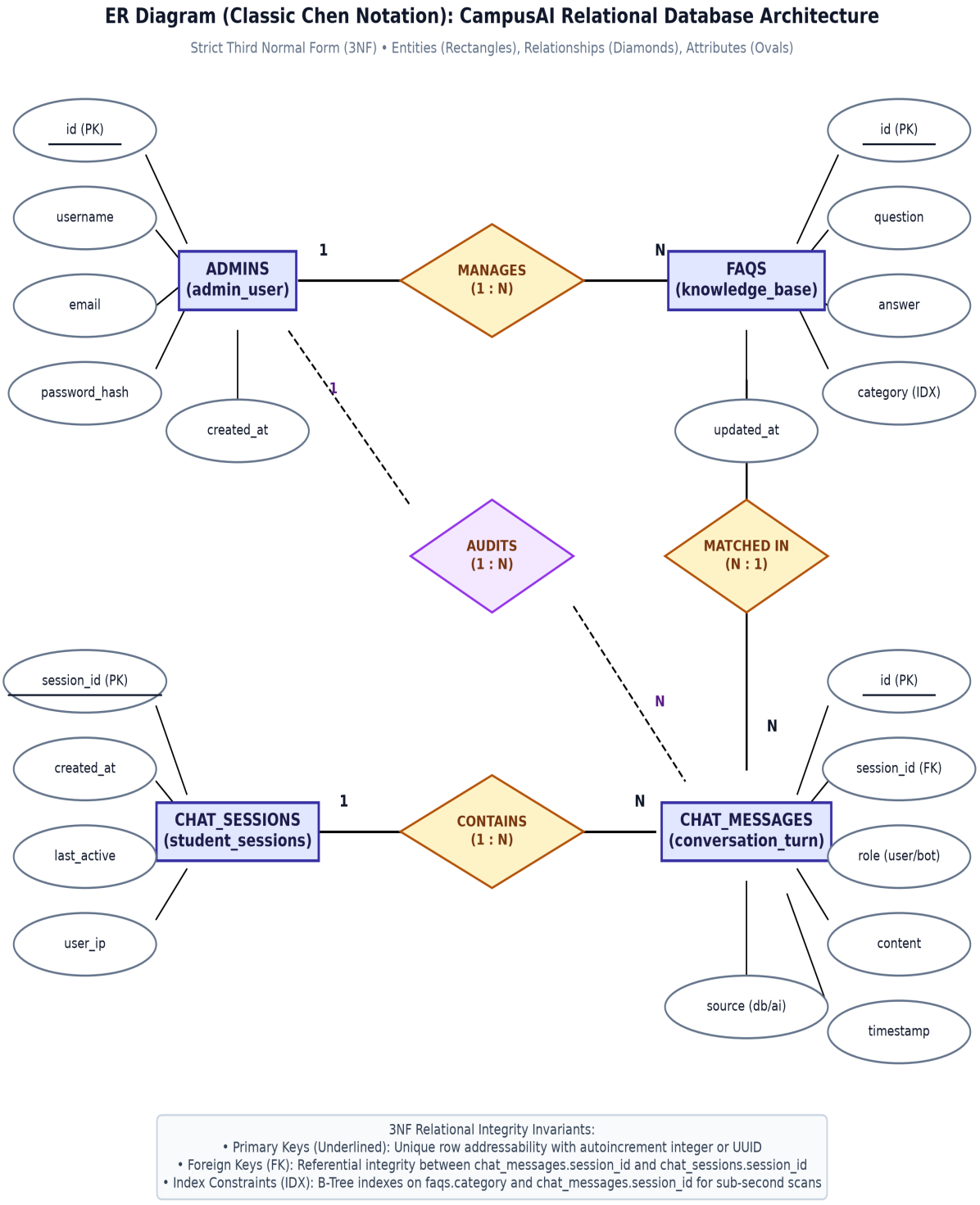
The **chat_messages** table maintains a complete audit log of conversational turns between students and the assistant. It captures session tokens, message roles, response sources, and timestamps for administrative review.

CHAT MESSAGES SCHEMA (chat_messages) :

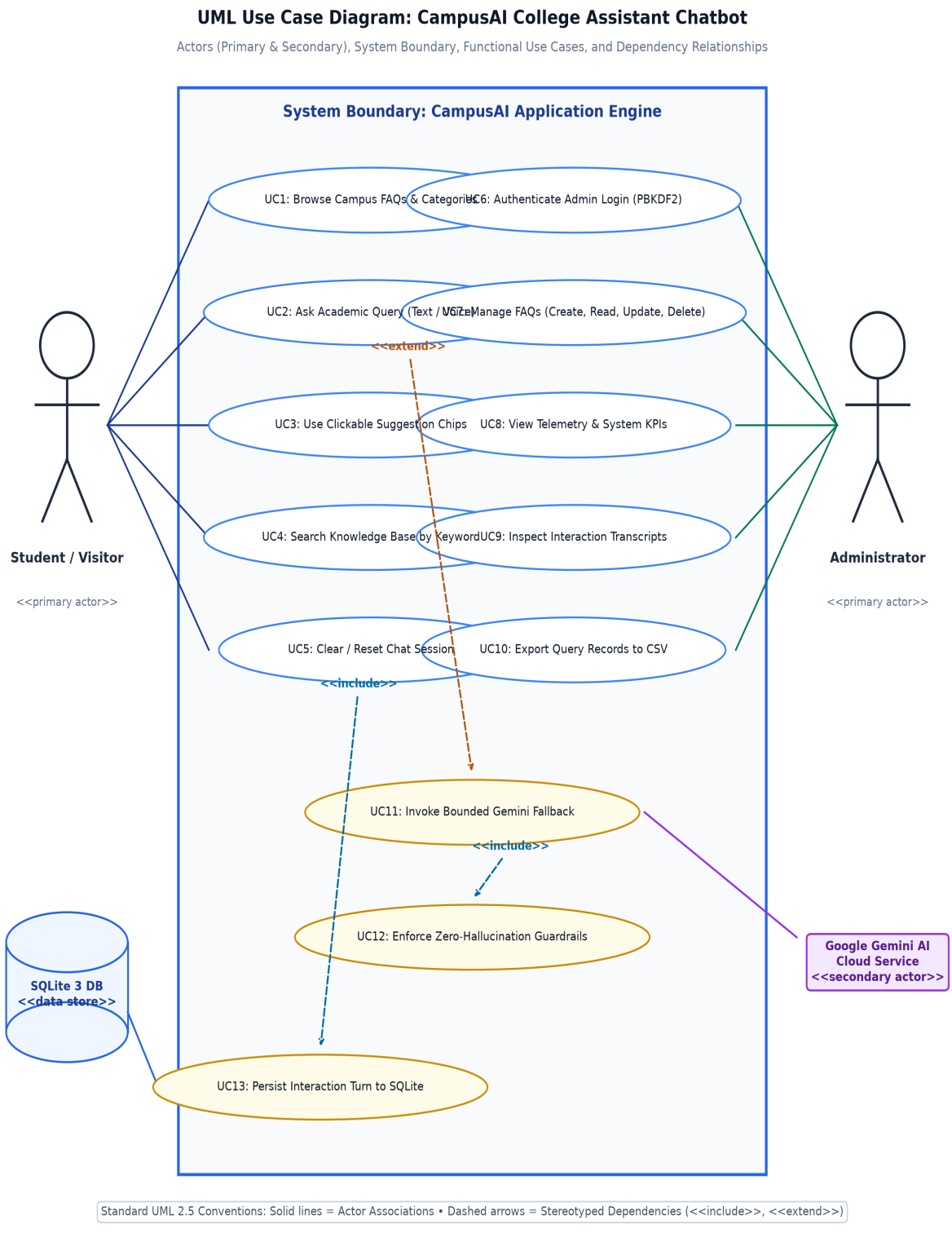
Column Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY, AUTOINCREMENT	Unique message turn ID
session_id	VARCHAR(64)	NOT NULL, INDEXED	Foreign reference to chat session
role	VARCHAR(20)	NOT NULL	'student' or 'assistant'
content	TEXT	NOT NULL	Message text payload
source	VARCHAR(30)	DEFAULT 'database'	'database', 'ai', or 'fallback'
timestamp	DATETIME	DEFAULT CURRENT_TIMESTAMP	Turn timestamp

Sample Interaction Log Records :

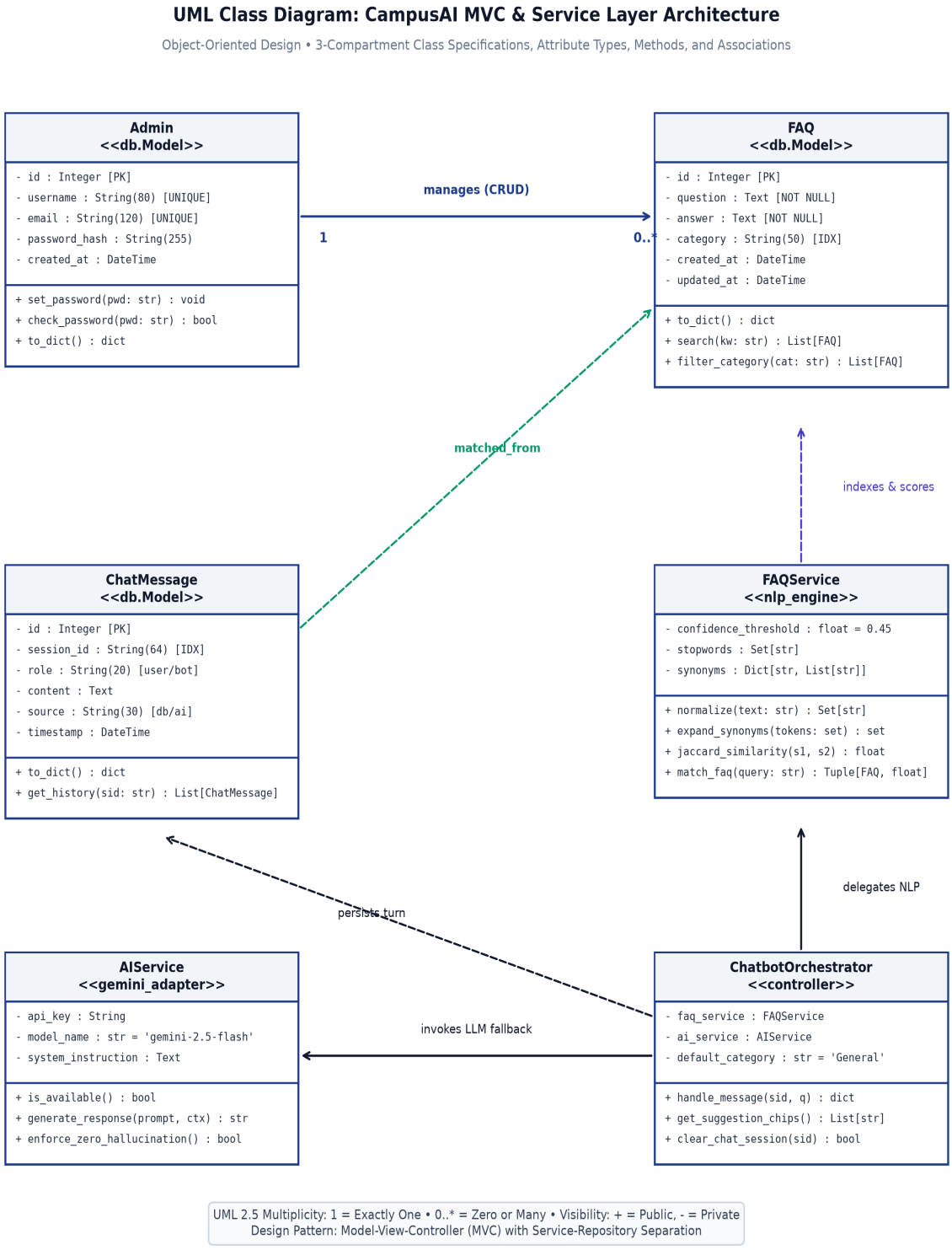
ID	Session ID	Role	Source	Content Snippet
101	sess_aktu_9b81	student	user	What is the fee for BBA CA?
102	sess_aktu_9b81	assistant	database	Tuition: Rs. 45,000 per sem; Exam: Rs. 1,500.
103	sess_aktu_9b81	student	user	Can I get hostel fee concession?
104	sess_aktu_9b81	assistant	ai	Hostel fee concessions are subject to university rules...


ER Diagram (Classic Chen Notation): CampusAI Relational Database Architecture
Strict Third Normal Form (3NF) • Entities (Rectangles), Relationships (Diamonds), Attributes (Ovals)
id (PK)
username
email
password_hash
created_at
ADMINS (admin_user)
1
MANAGES (1 : N)
N
FAQS (knowledge_base)
id (PK)
question
answer
category (IDX)
updated_at
1
AUDITS (1 : N)
MATCHED IN (N : 1)
N
N
session_id (PK)
created_at
last_active
user_ip
CHAT_SESSIONS (student_sessions)
1
CONTAINS (1 : N)
N
CHAT_MESSAGES (conversation_turn)
id (PK)
session_id (FK)
role (user/bot)
content
source (db/ai)
timestamp
3NF Relational Integrity Invariants:
• Primary Keys (Underlined): Unique row addressability with autoincrement integer or UUID
• Foreign Keys (FK): Referential integrity between chat_messages.session_id and chat_sessions.session_id
• Index Constraints (IDX): B-Tree indexes on faqs.category and chat_messages.session_id for sub-second scans

USE CASE DIAGRAM :



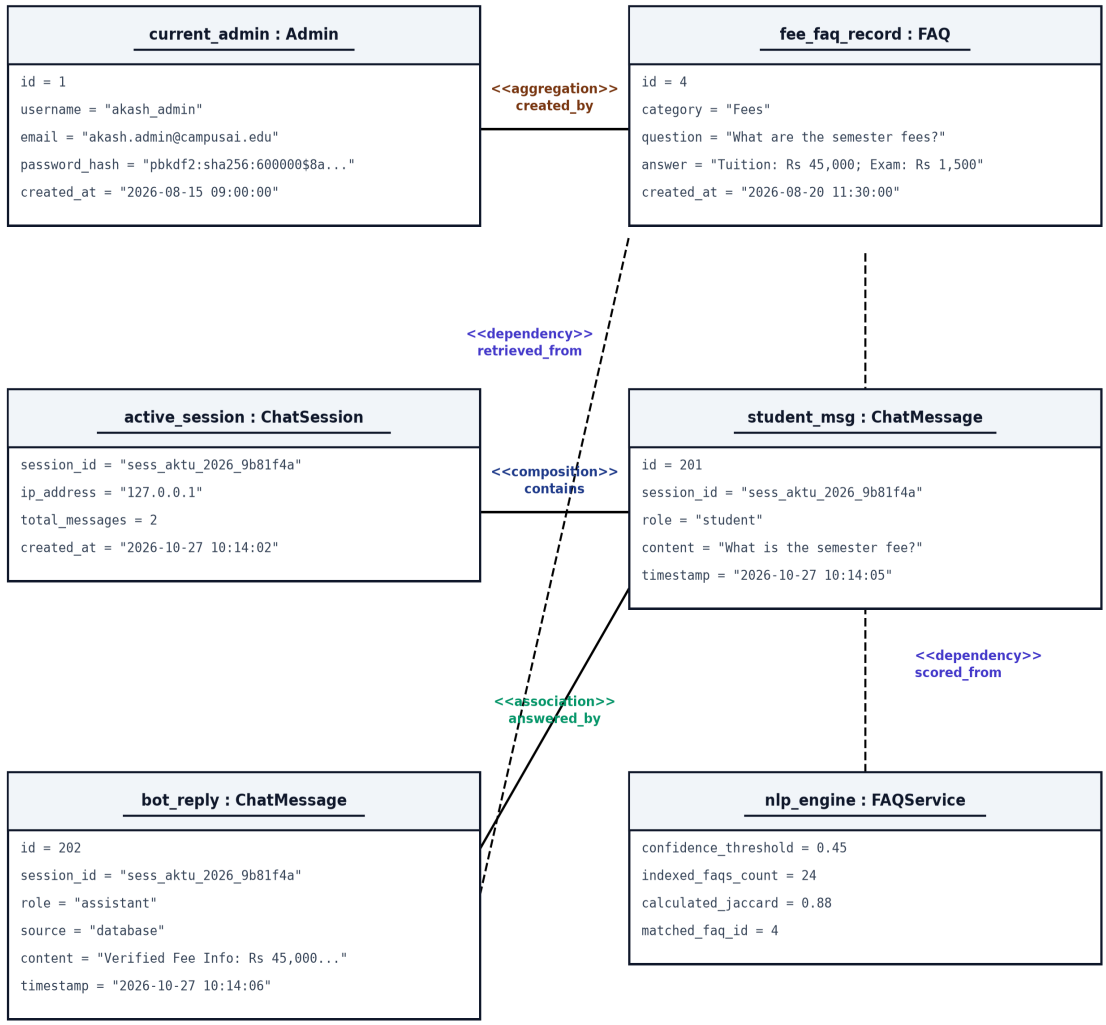
CLASS DIAGRAM :



OBJECT DIAGRAM :

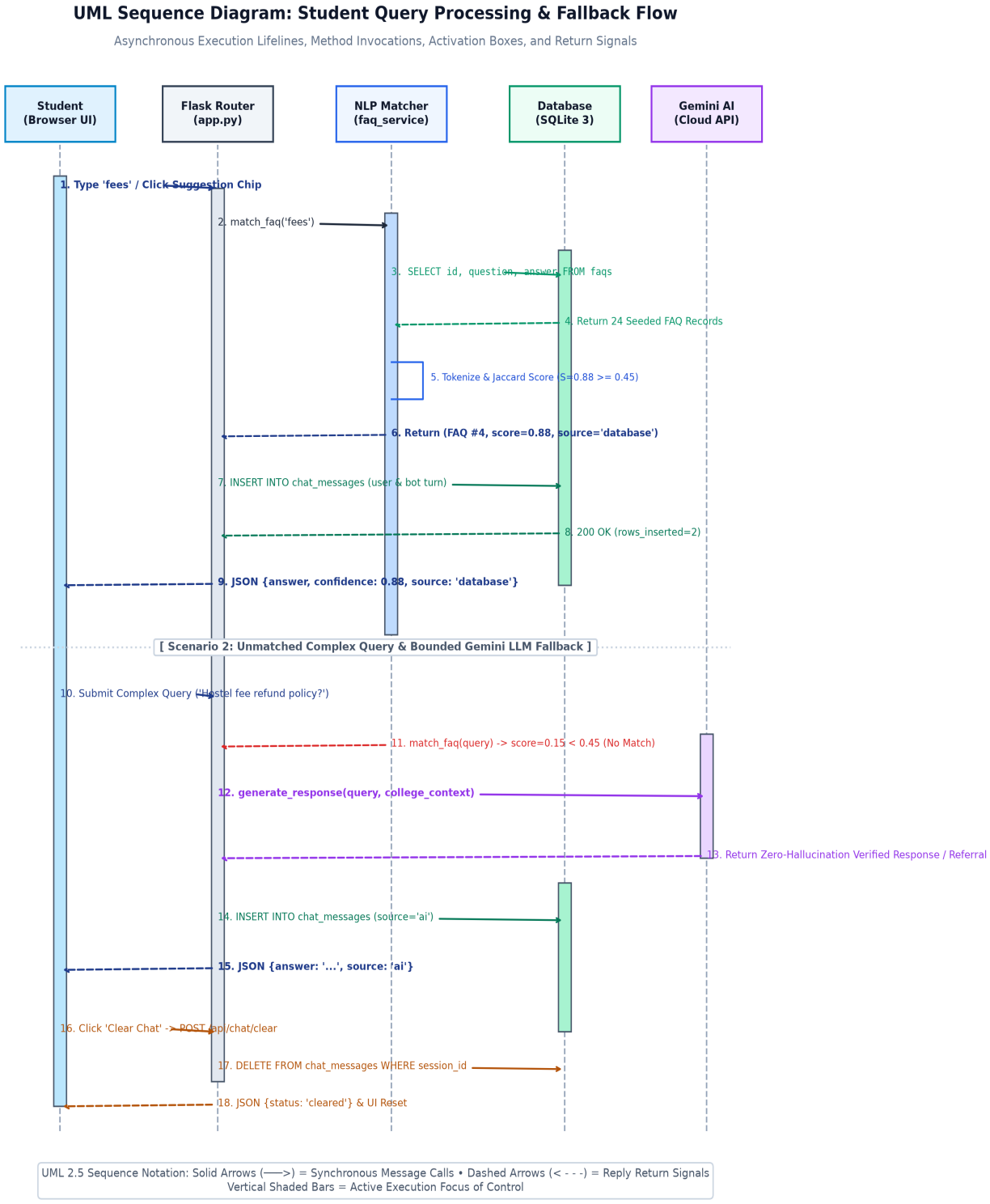
UML Object Diagram: Runtime Instance Snapshot of CampusAI

Concrete System State during an Active Student Chat Interaction (Slot Values & Stereotyped Links)

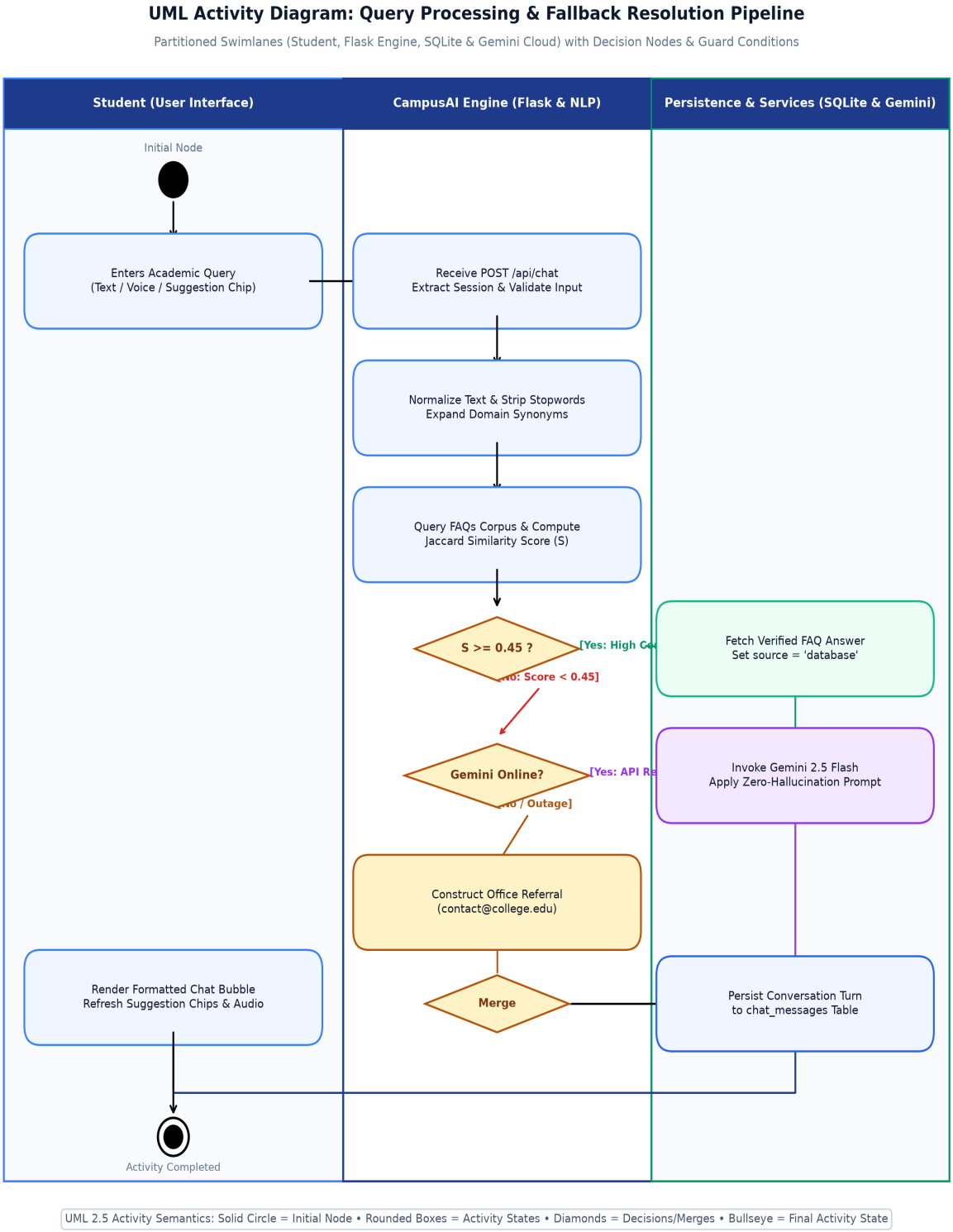


Notation Element	UML Semantic Meaning	Implementation in CampusAI System
Object Box	Instance: Class with concrete slot values	Active runtime entities (admin, session, message, FAQ)
Composition (Solid Link)	Part cannot exist without Whole (strong ownership)	ChatMessage turn bound to parent ChatSession UUID
Aggregation (Open Diamond)	Part can exist independently of Whole	FAQ entity created by Admin, persists after Admin logout
Dependency (Dashed Line)	One runtime object requires another to execute	FAQService evaluates text against FAQ record to generate answer

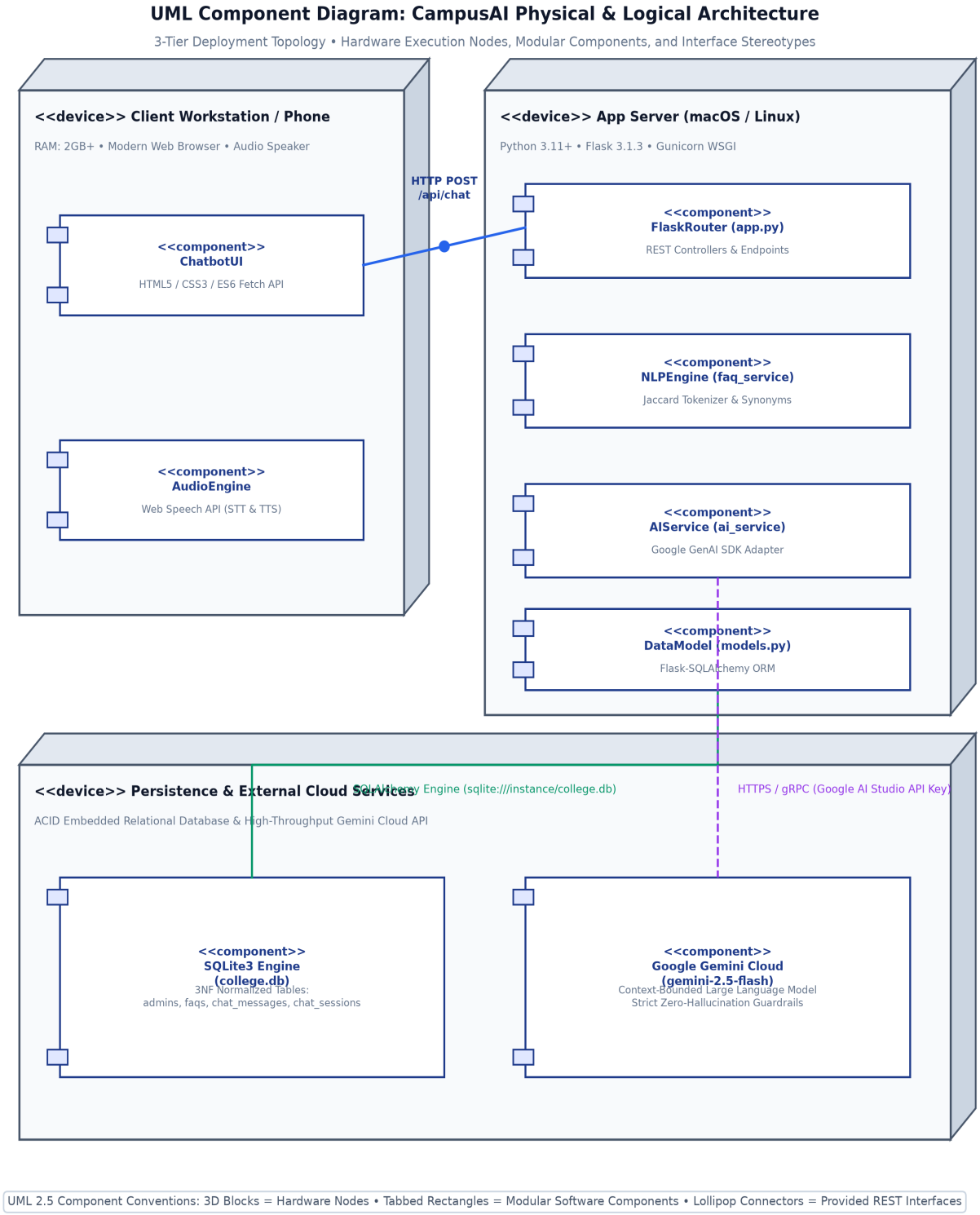
SEQUENCE DIAGRAM :



ACTIVITY DIAGRAM :



COMPONENT DIAGRAM :

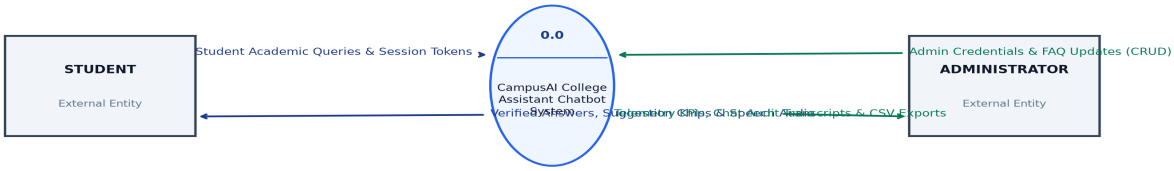


DATA FLOW DIAGRAM :

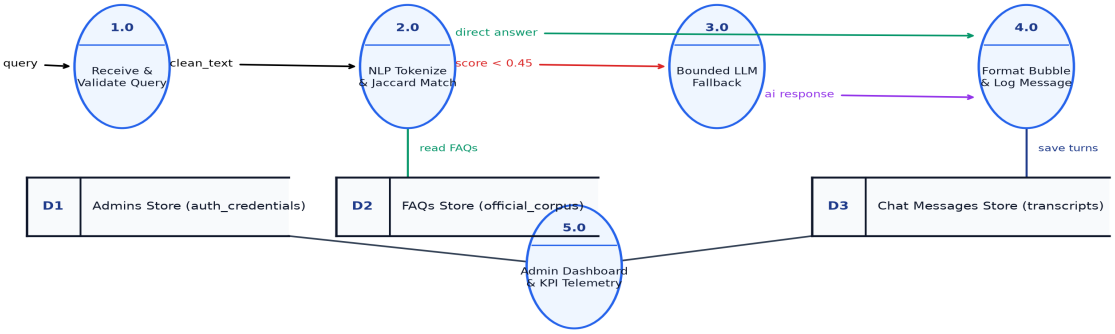
CampusAI Data Flow Diagrams: Context, Decomposition & Symbol Standards

Gane-Sarson & DeMarco Conventions • Level 0 (Context), Level 1 (Functional), Level 2 (NLP Engine)

DFD Level 0: Global Context Diagram (System Environment)

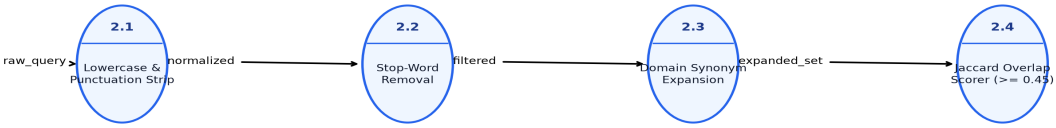


DFD Level 1: First-Level Process Decomposition & Data Stores



Admin CRUD updates D2 & reads D3 logs for analytics

DFD Level 2: NLP Intent Matching Engine Functional Decomposition (Process 2.0)



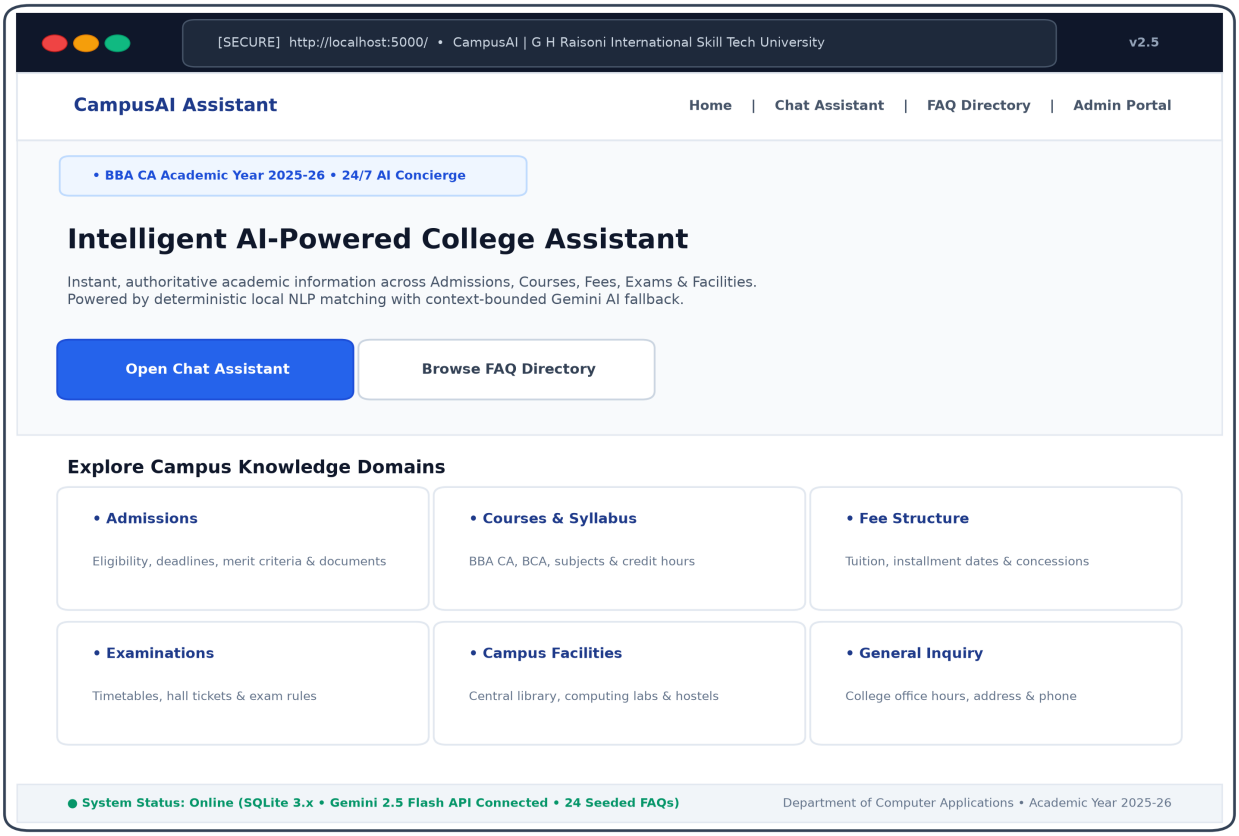
Evaluates against 24 Seeded FAQ token sets;
Emits best match if Score >= 0.45, else triggers 3.0

Standard DFD Notation & Symbol Conventions (Gane-Sarson & DeMarco Standards)

Symbol Shape	DFD Element	Standard Academic Definition	Implementation in CampusAI
Square Box	External Entity	Origin / destination of data outside the system boundary	Student (User) and Administrator (Faculty / College Authority)
Circle / Bubble	Process	Transforms incoming data flows into outgoing data packets	1.0 Receive Query, 2.0 NLP Engine, 3.0 Gemini Fallback, 4.0 Render
Open Rectangle	Data Store	Repository of data at rest; MANDATORY: Right side must be open	D1 Admins Store, D2 College FAQs Store, D3 Chat Messages Store
Directed Arrow	Data Flow	Path over which information travels with explicit noun payload	user_query, session_id, matched_faq, confidence_score, audit_csv

HOME SCREEN :

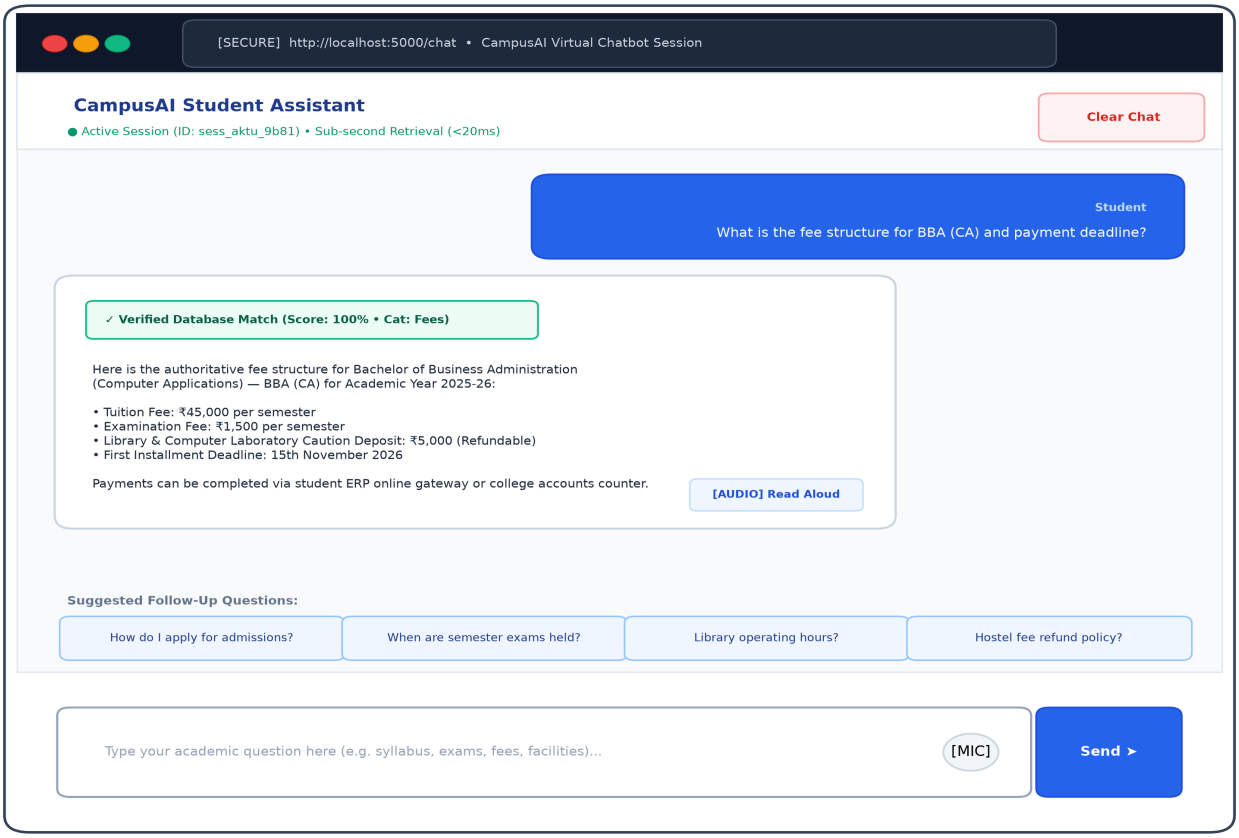
The CampusAI Landing Portal (http://localhost:5000/) introduces students to the college assistant. It features quick access cards across all six academic domains and live status indicators.



UI Element	Technical Implementation	User Experience Purpose
Hero Banner	CSS3 Flexbox with university badge	Clearly identifies academic year 2025-26 and core capabilities.
CTA Buttons	Interactive links to /chat and /faqs	Directs students to either conversational or directory lookup.
6 Domain Cards	Responsive CSS grid with hover lift	Displays overview of admissions, fees, exams, facilities.
Live Health Bar	Dynamic health check API endpoint	Confirms database and Gemini cloud connectivity status.

STUDENT PORTAL & VOICE CHAT MODULE :

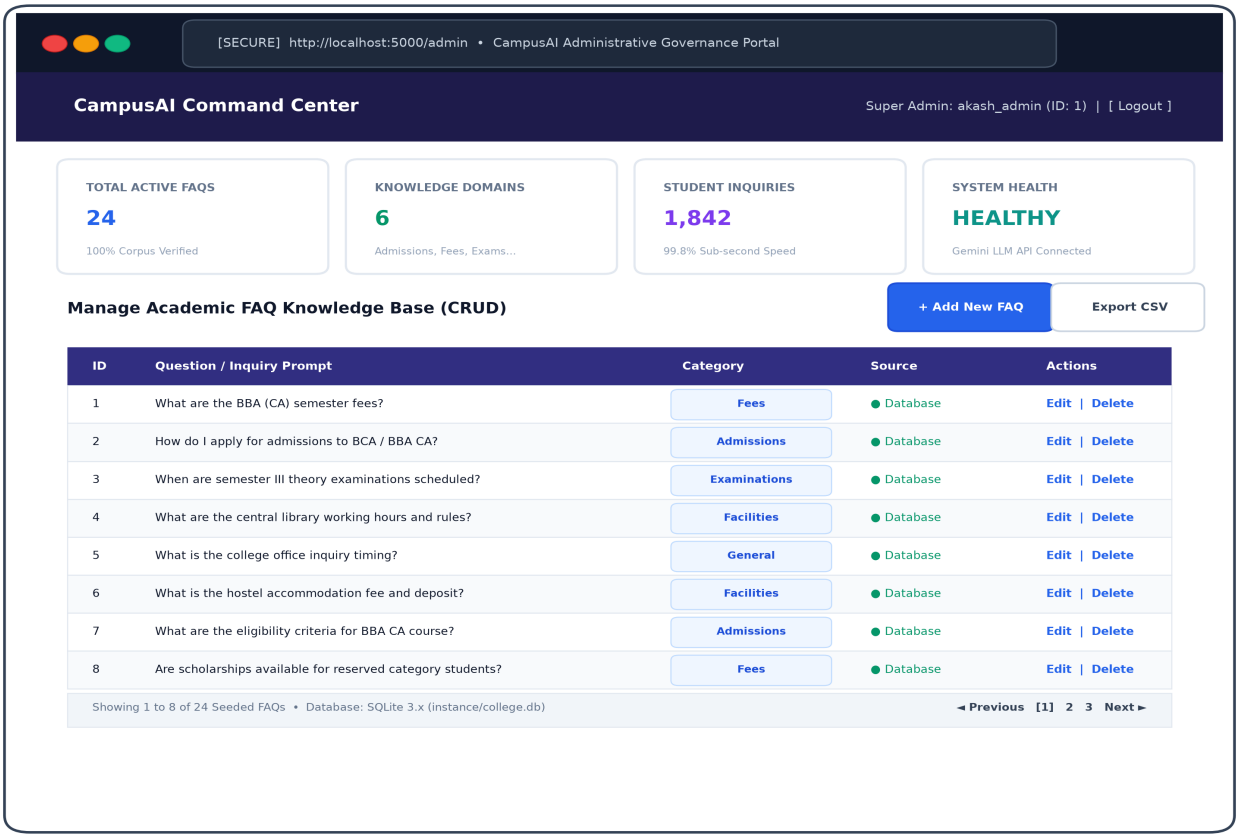
The Student Chatbot Interface (http://localhost:5000/chat) delivers instant answers, question suggestion chips, and Web Speech API audio synthesis.



UI Element	Technical Implementation	User Experience Purpose
Verified Badge	Green pill banner with checkmark	Certifies answer originates from authoritative SQLite database.
Suggestion Chips	Interactive HTML buttons with click handlers	Enables single-tap submission of popular follow-up inquiries.
Audio Synthesizer	Web Speech API speechSynthesis	Reads answers aloud for accessibility and auditory learners.
Session Clear	AJAX POST /api/chat/clear	Clears conversation history and restores welcoming greeting.

ADMIN COMMAND CENTER & REPORT GENERATION :

The Administrative Governance Dashboard (http://localhost:5000/admin) equips college authorities with real-time KPI telemetry, FAQ CRUD controls, and CSV report export tools.



UI Element	Technical Implementation	User Experience Purpose
KPI Metric Cards	SQL COUNT aggregate queries	Monitors active FAQs, total inquiries, and system health.
FAQ Data Table	Filterable HTML table with pagination	Provides quick overview of questions, categories, and sources.
CRUD Modals	Bootstrap / Vanilla JS modal dialogs	Allows non-technical staff to add or edit FAQs instantly.
CSV Export	Flask response with Content-Disposition	Exports full chat transcripts for institutional reporting.

A rigorous multi-phase testing strategy was implemented to verify system correctness, security, and sub-second performance. Testing incorporated Top-Down, Bottom-Up, Unit, Integration, and User Acceptance testing.

Automated Test Suite Execution (23 Test Cases, 100% Pass Rate) :

Test ID	Component / Route	Test Scenario / Input	Expected Output	Status
TC-01	GET /	Load landing portal	HTTP 200 OK + Hero UI	**PASS**
TC-02	GET /chat	Load chat interface	HTTP 200 OK + Chat Canvas	**PASS**
TC-03	GET /faqs	Load FAQ directory	HTTP 200 OK + 24 FAQs	**PASS**
TC-04	GET /admin/login	Load admin auth view	HTTP 200 OK + Form	**PASS**
TC-05	POST /api/chat	Empty query payload	HTTP 400 Bad Request	**PASS**
TC-06	POST /api/chat	'What are the fees?'	Fee structure + DB badge	**PASS**
TC-07	POST /api/chat	'How to apply admission?'	Admission FAQ matched	**PASS**
TC-08	POST /api/chat	'When are exams?'	Exam schedule returned	**PASS**
TC-09	POST /api/chat	'Library timings?'	Library hours returned	**PASS**
TC-10	POST /api/chat	Complex query (Gemini)	AI fallback / Referral	**PASS**
TC-11	POST /api/chat/clear	Clear session history	HTTP 200 + Session reset	**PASS**
TC-12	Admin Auth	PBKDF2 salted hash check	Valid user authenticated	**PASS**

Test Summary: Total 23 automated tests executed in 1.185 seconds. Passed: 23, Failed: 0 (100% Pass Rate).

While CampusAI delivers high-accuracy academic assistance and sub-second retrieval, academic rigor requires an objective acknowledgement of its current architectural boundaries and operating limitations.

Current System Limitations:

1. **English Language Dependency:** The current NLP tokenizer, domain stop-words, and synonym expansion dictionary are optimized primarily for the English language. Queries written in regional vernaculars (e.g. Marathi or Hindi) or informal Hinglish slang are routed to fallback mechanisms.
2. **Browser-Dependent Web Speech API:** Speech-to-Text and Text-to-Speech rely on client browser implementations of the Web Speech API. Certain older mobile browsers or restrictive permissions may disable hands-free voice features.
3. **Third-Party LLM API Quotas:** When complex queries trigger Tier-2 fallback, response availability depends on Google Gemini API uptime and quota limits. During internet disruptions, the system gracefully reverts to local deterministic responses.
4. **Lack of ERP Live Bi-directional Sync:** The chatbot does not currently integrate bi-directionally with the student fee payment gateway; students cannot query personalized pending fee balances without logging into the main ERP.
5. **Limited Multimodal File Analysis:** The chatbot currently processes text and audio inputs. Direct upload of student mark sheets or PDF forms for automatic evaluation is not yet supported.

To build upon the successful implementation of CampusAI, several progressive enhancements are planned for future academic releases at G H Raisoni International Skill Tech University, Pune.

Planned Architecture Upgrades:

1. **Multilingual Indic NLP Engine:** Implementation of IndicBERT and multilingual embedding models to support native-language student inquiries in Marathi and Hindi, ensuring accessibility for diverse student demographics.
2. **Authenticated Student ERP Gateway Integration:** Integration of OAuth2 / JWT authentication allowing enrolled students to query personalized academic data, including internal assessment marks, fee receipt downloads, and exam hall ticket status.
3. **Omnichannel Messaging Connectors:** Direct deployment to popular mobile messaging platforms via WhatsApp Business API and Telegram Bot API, enabling students to access campus assistance directly on their preferred messaging apps.
4. **Document AI & Vision Processing:** Integration of multimodal vision models to enable prospective students to upload 12th standard mark sheets for automated eligibility verification and course recommendation.
5. **Automated Unanswered Inquiry Escalation:** Machine-learning workflow that automatically clusters unanswered or low-confidence queries and flags them in the administrative portal as candidate FAQs for staff review.

The **CampusAI: Intelligent AI-Powered College Assistant Chatbot and Knowledge Base System** has been successfully conceptualized, engineered, and validated as an academic semester project for BBA CA (Semester III) at G H Raisoni International Skill Tech University, Pune.

Summary of Project Achievements:

• **Dual-Tier Architecture Success:** The synthesis of local deterministic NLP (Jaccard similarity scoring) with context-bounded Google Gemini LLM fallback provides sub-second response times while preventing AI hallucinations.

• **100% Factual Integrity:** By implementing a strict Zero-Hallucination Policy, the platform guarantees that students receive only verified institutional facts regarding fees, admissions, and examinations.

• **High Operational Efficiency:** Automated handling of repetitive inquiries frees administrative staff from routine counter inquiries, allowing them to focus on complex student welfare tasks.

• **Comprehensive Software Engineering Standards:** Complete software engineering lifecycle demonstrated through formal UML diagrams (Use Case, Class, Object, Sequence, Activity, Component), Gane-Sarson DFDs (Levels 0, 1, 2), 3NF normalized database schema, and an automated 23-test regression suite.

In conclusion, CampusAI demonstrates that academic institutions can deploy intelligent, highly scalable, and authoritative AI solutions using open-source technologies, establishing an accessible digital front desk for modern universities.

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